

Sample Papers

MTH501 Linear Algebra

Mid Term Examination – Spring 2006

Time Allowed: 90 Minutes

Please read the following instructions carefully before attempting any of the questions:

1. Attempt all questions. Marks are written adjacent to each question.
2. Do not ask any questions about the contents of this examination from anyone.
 - a. If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.
 - b. If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.
 - c. Write all steps, missing steps may lead to deduction of marks.

****WARNING: Please note that Virtual University takes serious note of unfair means. Anyone found involved in cheating will get an 'F' grade in this course.**

Question No. 1

Marks : 8

The augmented matrix of a linear system has been transformed by row operations into the form below. Write the general solution in **Parametric vector form**.

$$\begin{bmatrix} 1 & 5 & 2 & -6 \\ 0 & 4 & -7 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Question No. 2

Marks : 8

A transformation $T: R^2 \rightarrow R^3$ defined by $T(x) = Ax$, so that

$$T(x) = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -x_1 + 7x_2 \end{bmatrix}$$

Question No. 3

Marks : 5

Calculate the **area of the parallelogram** determined by the points $(-2, -2)$, $(0, 3)$, $(4, -1)$ and $(6, 4)$.

Question No. 4

Marks : 1

A linear transformation $T: R^2 \rightarrow R^2$ defined by $T(x) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ then $T\left(\begin{bmatrix} 4 \\ 1 \end{bmatrix}\right) =$ _____

★ $\begin{bmatrix} -1 \\ 4 \end{bmatrix}$

★ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

★ $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$

★ $\begin{bmatrix} -3 \\ 3 \end{bmatrix}$

Question No. 5

Marks : 6

Let $A = \begin{bmatrix} -6 & 12 \\ -3 & 6 \end{bmatrix}$ and $w = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ Determine if w is in $\text{Col } A$. Is w in $\text{Null } A$?

Question No. 6

Marks : 1

$$A = \begin{bmatrix} 1 & -2 & 5 & 0 \\ 0 & 4 & 4 & -1 \\ 0 & 0 & 5 & 7 \\ 0 & 0 & 0 & 7 \end{bmatrix} \quad \det A = 140 \quad \text{Then } \det A^T = \text{-----}$$

- ★ 0
- ★ 140
- ★ 6
- ★ -20

Question No. 7

Marks : 1

The augmented matrix of system of linear equations $\begin{matrix} x_1 + 2x_3 = 0 \\ x_1 + x_2 - 9x_3 = -1 \end{matrix}$ is

★ $\begin{bmatrix} 1 & 0 & 2 & 0 \\ 1 & 1 & -9 & -1 \end{bmatrix}$

★ $\begin{bmatrix} 1 & 2 & 0 & 0 \\ 1 & 1 & -9 & 1 \end{bmatrix}$

★ $\begin{bmatrix} 1 & 2 & 0 \\ 1 & 1 & -9 \end{bmatrix}$

★ None of the other.

Question No. 8

Marks : 1

Identify the true statement

- ★ The columns of a matrix A is linearly independent if the equation $Ax = \theta$ has trivial solution
- ★ The columns of a matrix A is linearly independent if the equation $Ax = \theta$ has non trivial solution.
- ★ If one row in echelon form of an augmented matrix is $[0 \ 0 \ 0 \ 0 \ 5 \ 0]$, then associated linear system is inconsistent.
- ★ Every transformation is linear transformation.

Question No. 9

Marks : 8

Find **LU factorization** of the matrix $A = \begin{bmatrix} 2 & 5 \\ -3 & -4 \end{bmatrix}$

Question No. 10

Marks : 1

Let $T: R^n \rightarrow R^m$ be a linear transformation and A be the standard matrix of T , then T maps

R^n onto R^m if and only if

- ★ The columns of A span R^m .
- ★ The columns of A span R^n
- ★ The columns of A are linearly independent
- ★ None of the other.